

Da Extração de Características à Combinação de Classificadores

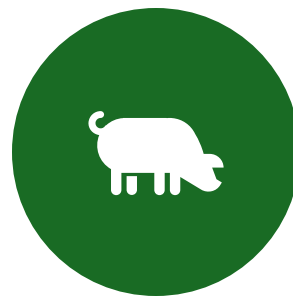
Uma Análise Supervisionada de Imagens de Cães e Gatos

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Introdução



Problema: classificação binária de cães e gatos.



Abordagem supervisionada com HOG, LBP e classificadores base (k-NN, DT, NB, MLP).

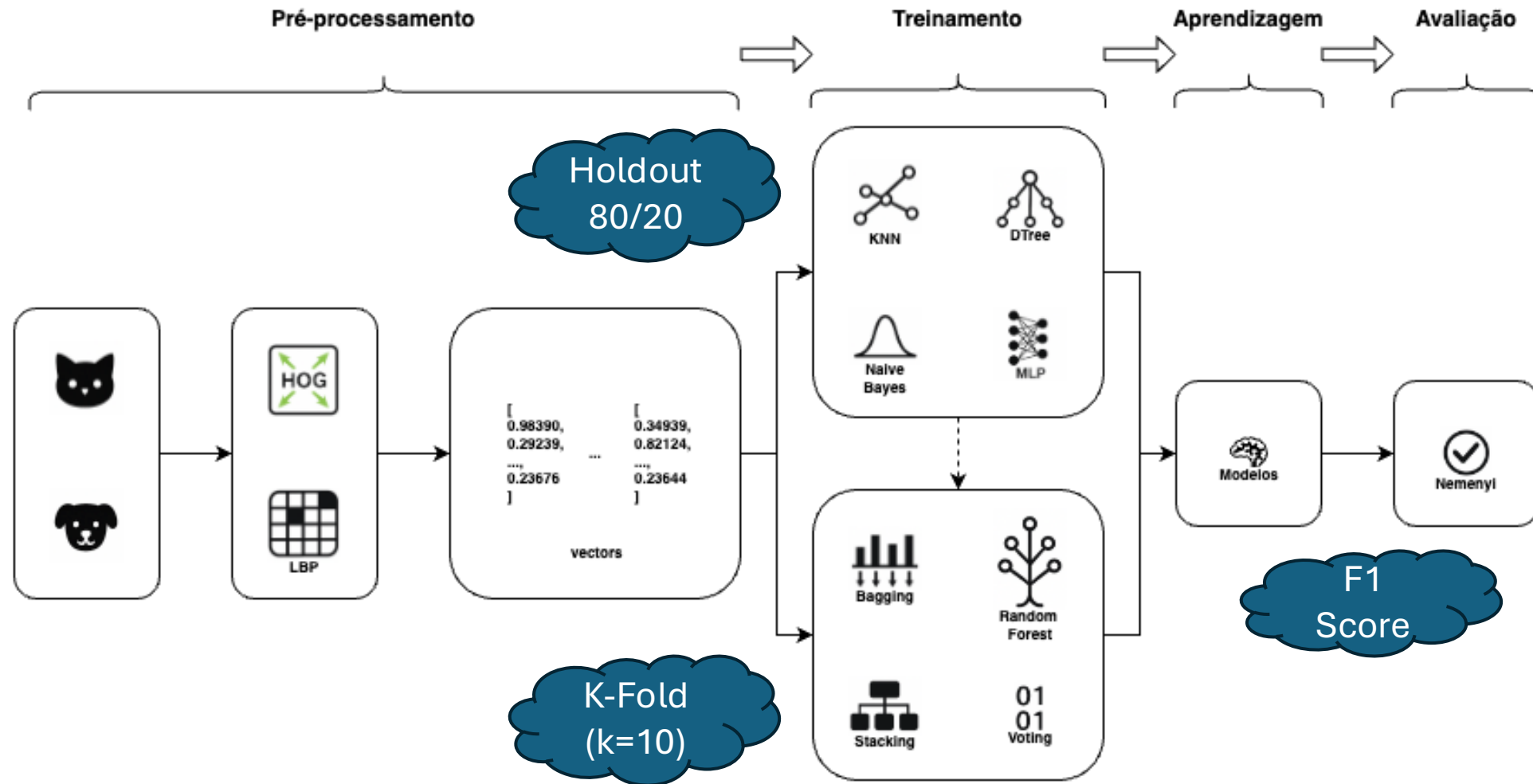


Comitês aplicados: Bagging, Random Forest, Voting e Stacking.



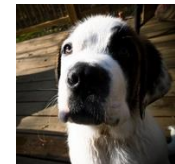
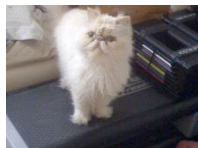
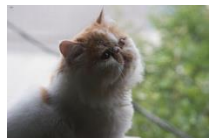
Objetivo: comparar desempenho e generalização.

Protocolo e Metodologia





Base de Dados



- 800 imagens (400 cães / 400 gatos).
- Raças: Basset Hound, Saint Bernard, Birman, Persian.
- Variabilidade: iluminação, fundo e foco.
- Exemplo visual de cada classe.



Pré-processamento



[
0.98390,
0.29239,
...,
0.23676
]

Nº	Dataset	(linhas, colunas)	Características da extração	Tipo de Treinamento	Média	Desvio Padrão
1	hogfeat_128_16_4_9_pca	(800, 103)	128/16/4/9/pca	Holdout (90/10) kFold (n=10)	0,752141	0,020340
2	hogfeat_256_64_2_18	(800, 648)	256/64/2/18	Holdout (90/10) kFold (n=10)	0,725717	0,026902
3	hogfeat_256_64_2_9	(800, 324)	256/64/2/9	Holdout (90/10) kFold (n=10)	0,720167	0,025322
4	hogfeat_128_16_4_9	(800, 3600)	128/16/4/9	Holdout (90/10) kFold (n=10)	0,719345	0,042608
5	hogfeat_256_32_2_9_pca	(800, 93)	256/32/2/9/pca	Holdout (90/10) kFold (n=10)	0,714259	0,027824
6	hogfeat_128_16_2_9_pca	(800, 117)	128/16/2/9/pca	Holdout (90/10) kFold (n=10)	0,704402	0,022513
7	hogfeat_128_32_2_9	(800, 324)	128/32/2/9	Holdout (90/10) kFold (n=10)	0,693092	0,048442
8	lbpfeat_256_6_48	(800, 50)	256/6/48	Holdout (90/10) kFold (n=10)	0,673671	0,037790
9	lbpfeat_256_12_96	(800, 98)	256/12/96	Holdout (90/10) kFold (n=10)	0,669408	0,025862
10	hogfeat_256_32_2_9	(800, 1764)	256/32/2/9	Holdout (90/10) kFold (n=10)	0,668029	0,044995

- Variabilidade de atributos e características do extrator
- Identificar melhor extrator/configurações
- Aplicação PCA a 95%

Nº	Dataset	(linhas, colunas)	Características da extração	Tipo de Treinamento	Média	Desvio Padrão
11	hogfeat_128_16_2_9	(800, 1764)	128/16/2/9	Holdout (90/10) kFold (n=10)	0,654806	0,024189
12	lbpfeat_256_24_192	(800, 194)	256/24/192	Holdout (90/10) kFold (n=10)	0,643985	0,043010
13	lbpfeat_256_3_24	(800, 26)	256/3/24	Holdout (90/10) kFold (n=10)	0,640101	0,041996
14	hogfeat_128_8_2_9	(800, 8100)	128/8/2/9	Holdout (90/10) kFold (n=10)	0,565446	0,040677
15	lbpfeat_256_1_8	(800, 10)	256/1/8	Holdout (90/10) kFold (n=10)	0,565417	0,037681
16	hogfeat_128_8_2_9_pca	(800, 287)	128/8/2/9/pca	Holdout (90/10) kFold (n=10)	0,555172	0,046436
17	hogfeat_256_4_2_9	(800, 142884)	256/4/2/9	Holdout (90/10) kFold (n=10)	0,543699	0,038550
18	hogfeat_256_8_2_9	(800, 34596)	256/8/2/9	Holdout (90/10) kFold (n=10)	0,527632	0,040768
19	hogfeat_256_8_2_9_pca	(800, 394)	256/8/2/9/pca	Holdout (90/10) kFold (n=10)	0,514819	0,043531
20	hogfeat_256_4_2_9_pca	(800, 507)	256/4/2/9/pca	Holdout (90/10) kFold (n=10)	0,457396	0,053933

LBP: Resolução/Raio/Pontos

HOG: Resolução/Pixels per Cell/Cells per Block/Orientations/PCA

K-NN



•GridSearch:

```
'n_neighbors': [1, 3, 5, 7, 9, 11, 13, 15],  
'metric': ['euclidean', 'manhattan', 'chebyshev'],  
'weights': ['uniform', 'distance']
```

Friedman_P_Valor

0,6153355838888000

config_rank	params
1	{'metric': 'chebyshev', 'n_neighbors': 3, 'weights': 'distance'}
2	{'metric': 'chebyshev', 'n_neighbors': 3, 'weights': 'uniform'}
3	{'metric': 'chebyshev', 'n_neighbors': 5, 'weights': 'distance'}
4	{'metric': 'chebyshev', 'n_neighbors': 5, 'weights': 'uniform'}
5	{'metric': 'chebyshev', 'n_neighbors': 11, 'weights': 'distance'}
6	{'metric': 'chebyshev', 'n_neighbors': 11, 'weights': 'uniform'}
7	{'metric': 'chebyshev', 'n_neighbors': 15, 'weights': 'distance'}
8	{'metric': 'chebyshev', 'n_neighbors': 15, 'weights': 'uniform'}
9	{'metric': 'chebyshev', 'n_neighbors': 7, 'weights': 'distance'}
10	{'metric': 'chebyshev', 'n_neighbors': 7, 'weights': 'uniform'}

config_rank	Média de f1_score	DesvPad de f1_score	Média de execution_time	DesvPad de execution_time
2	0,6086909267387500	0,0891748838700589	0,277	0,54
1	0,6073320740422070	0,0883254523171088	0,277	0,53
5	0,6046680669636470	0,1145490784145480	0,277	0,53
6	0,6037675239603530	0,1138986397206530	0,277	0,53
7	0,6031921862310220	0,1187065619401660	0,281	0,54
8	0,6019436207301170	0,1178022587006150	0,279	0,54
9	0,6006591314461030	0,1047153241950820	0,281	0,55
3	0,6003713458691610	0,0977810894414382	0,276	0,53
10	0,6002950206491470	0,1044919617264860	0,276	0,53
4	0,6001952491936030	0,0977942368311956	0,274	0,53



Decision Tree

•GridSearch:

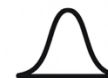
```
'criterion': ['gini', 'entropy', 'log_loss'],  
'max_depth': [2,3,4,5,6,7,8,9,10,11,12,13,14,15]
```

Friedman_P_Valor
0,7160981986368600

config_rank	params
1	{'criterion': 'entropy', 'max_depth': 3}
2	{'criterion': 'log_loss', 'max_depth': 3}
3	{'criterion': 'gini', 'max_depth': 3}
4	{'criterion': 'entropy', 'max_depth': 5}
5	{'criterion': 'log_loss', 'max_depth': 4}
6	{'criterion': 'entropy', 'max_depth': 4}
7	{'criterion': 'gini', 'max_depth': 4}
8	{'criterion': 'log_loss', 'max_depth': 5}
9	{'criterion': 'entropy', 'max_depth': 10}
10	{'criterion': 'log_loss', 'max_depth': 9}

config_rank	Média de f1_score	DesvPad de f1_score	Média de execution_time	DesvPad de execution_time
8	0,676379	0,03	2,222	4,77
4	0,676242	0,03	2,252	4,87
7	0,676075	0,03	1,459	3,11
3	0,674964	0,04	1,139	2,42
1	0,674054	0,04	1,494	3,18
2	0,673509	0,04	1,491	3,19
5	0,672510	0,03	1,886	4,04
6	0,671537	0,03	1,896	4,05
10	0,664883	0,03	2,848	6,13
9	0,660403	0,03	2,899	6,25

Naive Bayes



•GridSearch:

```
'class': [GaussianNB, MultinomialNB, ComplementNB]
```

config_rank	params
1	{'nb_type': 'ComplementNB'}
2	{'nb_type': 'GaussianNB'}
3	{'nb_type': 'MultinomialNB'}

config_rank	Média de f1_score	DesvPad de f1_score	Média de execution_time	DesvPad de execution_time
2	0,737476	0,03	0,077	0,14
1	0,695284	0,10	0,058	0,09
3	0,649973	0,12	0,064	0,10

Friedman_P_Valor

0,0000056987208693

	config_1	config_2	config_3
config_1	1,0000000000000000	0,0545006301142829	0,0309911625911873
config_2	0,0545006301142829	1,0000000000000000	0,0000039727329548
config_3	0,0309911625911873	0,0000039727329548	1,0000000000000000



Rede Neural MLP

•GridSearch:

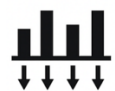
```
'hidden_layer_sizes': [(50), (100), (150), (100, 50), (150, 75), (200, 100), (200, 75, 25), (200, 100, 50) ],  
'activation': ['identity', 'logistic', 'tanh', 'relu'],  
'solver': ['adam', 'sgd'],  
'learning_rate_init': [0.001, 0.005, 0.01],
```

config_10
0,9999999999999770
0,0070074238418840
0,8919827960942220
0,0140392023329433
0,9180233004941860
0,1319771202779860
0,0744677090909467
0,9011778596711110
0,9011778596711110
1,0000000000000000

Friedman_P_Valor
0,0001937073329703

config_rank	params
1	{'activation': 'relu', 'hidden_layer_sizes': (200, 100), 'learning_rate_init': 0.005, 'max_iter': 500, 'solver': 'adam'}
2	{'activation': 'logistic', 'hidden_layer_sizes': (200, 100, 50), 'learning_rate_init': 0.005, 'max_iter': 500, 'solver': 'adam'}
3	{'activation': 'relu', 'hidden_layer_sizes': 50, 'learning_rate_init': 0.01, 'max_iter': 500, 'solver': 'adam'}
4	{'activation': 'tanh', 'hidden_layer_sizes': (150, 75), 'learning_rate_init': 0.01, 'max_iter': 500, 'solver': 'adam'}
5	{'activation': 'relu', 'hidden_layer_sizes': (200, 100), 'learning_rate_init': 0.01, 'max_iter': 500, 'solver': 'adam'}
6	{'activation': 'relu', 'hidden_layer_sizes': (200, 100, 50), 'learning_rate_init': 0.01, 'max_iter': 500, 'solver': 'sgd'}
7	{'activation': 'identity', 'hidden_layer_sizes': (200, 100, 50), 'learning_rate_init': 0.001, 'max_iter': 500, 'solver': 'adam'}
8	{'activation': 'relu', 'hidden_layer_sizes': (200, 100, 50), 'learning_rate_init': 0.005, 'max_iter': 500, 'solver': 'adam'}
9	{'activation': 'relu', 'hidden_layer_sizes': 150, 'learning_rate_init': 0.01, 'max_iter': 500, 'solver': 'adam'}
10	{'activation': 'relu', 'hidden_layer_sizes': (200, 100), 'learning_rate_init': 0.001, 'max_iter': 1500, 'solver': 'adam'}

config_rank	Média de f1_score	DesvPad de f1_score	Média de execution_time	DesvPad de execution_time
10	0,7650328505118750	0,0230171562074002	13,344	21,50
1	0,7632066363275250	0,0235917805211502	8,583	15,54
5	0,7590800878445580	0,0237553780875046	7,432	14,31
9	0,7563215895228020	0,0224701775342486	10,590	24,07
3	0,7538370282109150	0,0229801190190982	4,414	10,02
8	0,7530121803821610	0,0315290542425867	6,868	11,86
7	0,7509392469886280	0,0201726228055640	7,410	14,87
6	0,7441590467031480	0,0327506191334780	14,557	26,54
4	0,6799421622958330	0,1736803731157850	5,985	17,81
2	0,6735739089261490	0,1679990582117040	10,236	18,70



Bagging e Random Forest



config_rank	params
1	{'estimatorModel': 'knn', 'n_estimators': 10}
2	{'estimatorModel': 'knn', 'n_estimators': 20}
3	{'estimatorModel': 'knn', 'n_estimators': 30}
4	{'estimatorModel': 'dtree', 'n_estimators': 10}
5	{'estimatorModel': 'dtree', 'n_estimators': 20}
6	{'estimatorModel': 'dtree', 'n_estimators': 30}
7	{'estimatorModel': 'nb', 'n_estimators': 10}
8	{'estimatorModel': 'nb', 'n_estimators': 20}
9	{'estimatorModel': 'nb', 'n_estimators': 30}
10	{'estimatorModel': 'mlp', 'n_estimators': 10}
11	{'estimatorModel': 'mlp', 'n_estimators': 20}
12	{'estimatorModel': 'mlp', 'n_estimators': 30}

config_rank	Média de f1_score	DesvPad de f1_score	Média de execution_time	DesvPad de execution_time
12	0,7719039779956610	0,0180937012554618	177,201	198,71
10	0,7712791840815720	0,0200909953512338	57,692	64,39
11	0,7696792566156110	0,0180819220820803	117,835	132,88
6	0,7458714748810980	0,0259484040115356	4,735	9,66
5	0,7442152358315790	0,0222437702648676	3,216	6,50
9	0,7377738499001900	0,0306654417826667	0,765	0,89
7	0,7361618094517730	0,0302919122439861	0,359	0,40
8	0,7361351166944120	0,0318474285301911	0,565	0,65
4	0,7346079912850380	0,0274560198419171	1,686	3,29
2	0,6112579375630100	0,0891393753654913	1,990	3,08
3	0,6059815273030590	0,0916335845622388	2,859	4,57
1	0,6033509607702260	0,0876680720742989	0,974	1,47
Total Geral	0,7140181935311020	0,0801059564914686	30,823	89,68

config_rank	params
1	{'criterion': 'gini', 'n_estimators': 10}
2	{'criterion': 'gini', 'n_estimators': 20}
3	{'criterion': 'gini', 'n_estimators': 30}
4	{'criterion': 'gini', 'n_estimators': 100}
5	{'criterion': 'entropy', 'n_estimators': 10}
6	{'criterion': 'entropy', 'n_estimators': 20}
7	{'criterion': 'entropy', 'n_estimators': 30}
8	{'criterion': 'entropy', 'n_estimators': 100}
9	{'criterion': 'log_loss', 'n_estimators': 10}
10	{'criterion': 'log_loss', 'n_estimators': 20}
11	{'criterion': 'log_loss', 'n_estimators': 30}
12	{'criterion': 'log_loss', 'n_estimators': 100}

config_rank	Média de f1_score	DesvPad de f1_score	Média de execution_time	DesvPad de execution_time
12	0,7659989957775850	0,0241750995046156	1,106	1,10
5	0,7638341164473550	0,0228644554593418	1,112	1,10
11	0,7622043414869680	0,0214674490363658	1,111	1,10
8	0,7618075130661020	0,0270143929169610	1,112	1,10
9	0,7603339010833280	0,0211075858961711	1,115	1,10
6	0,7600096849220510	0,0220912189065318	1,112	1,10
10	0,7595816280948990	0,0233015426837849	1,110	1,10
7	0,7588287891710650	0,0231209857704459	1,106	1,09
2	0,7581311061092830	0,0203939655816427	1,007	0,95
1	0,7548653695319120	0,0189827044111651	1,005	0,95
3	0,7537548646831720	0,0233945949873007	1,009	0,95
4	0,7501009954842070	0,0245490587782699	1,003	0,95
Total Geral	0,7591209421548270	0,0227577688447900	1,076	1,03



Voting e Stacking

'estimatorConfigPeak': [

```
{'knn': 1 },
{'dtree': 1 },
{'knn': 6 },
{'mlp': 1 },
{'mlp': 6 },
```

```
{'knn': 2 },
{'dtree': 2 },
{'knn': 8 },
{'mlp': 2 },
{'dtree': 6 },
```

```
{'knn': 3 },
{'dtree': 3 },
{'knn': 7 },
{'mlp': 3 },
{'dtree': 4 },

{'knn': 4 },
{'knn': 5 },
{'dtree': 5 },
{'mlp': 4 },
{'mlp': 5 },
],
```

model	Voting
config_rank	params
	1 {'n_estimators': 5}
	2 {'n_estimators': 10}
	3 {'n_estimators': 15}
	4 {'n_estimators': 20}

config_rank	Média de f1_score	DesvPad de f1_score	Média de execution_time	DesvPad de execution_time
1	0,7584746877030580	0,0272835480459580	11,837	13,55
4	0,7532899043949790	0,0256100538609706	26,284	33,41
3	0,7491075627974190	0,0285417813779000	18,863	22,49
2	0,7449411213555900	0,0250620849054570	15,260	17,75
Total Geral	0,7514533190627610	0,0267155000068456	18,061	23,29

config_rank	params
	1 {'n_estimators': 5}
	2 {'n_estimators': 10}
	3 {'n_estimators': 15}
	4 {'n_estimators': 20}

config_rank	Média de f1_score	DesvPad de f1_score	Média de execution_time	DesvPad de execution_time
2	0,7688072833244000	0,0227414153067549	75,010	89,05
1	0,76742797444446680	0,0253534372739601	53,367	60,54
4	0,7666170883606170	0,0224617391138081	134,535	174,75
3	0,7657057701819960	0,0239152282917947	94,119	114,21
Total Geral	0,7671395290779200	0,0232973184344670	89,258	119,41

Comparativo Classificadores Base

dataset	shape	training_type	KNN Config 2	Dtree Config 8	NB Config 2	MLP Config 10
hogfeat_128_16_2_9	(800, 1765)	crossvalidation	0,4846328414620430	0,6355919193659980	0,7787775607163990	0,7835445083251580
		holdout	0,4739207508944350	0,6005050505050500	0,7752840909090900	0,7370833333333330
hogfeat_128_16_2_9_pca	(800, 118)	crossvalidation	0,7060158085674000	0,6878073106259910	0,6808356417552070	0,7410149121338570
		holdout	0,6731297953964190	0,6873133076019460	0,7071352393252710	0,7189389000829680
hogfeat_128_16_4_9	(800, 3601)	crossvalidation	0,5045323437290420	0,6548173044202140	0,7349197981599640	0,7755239593550670
		holdout	0,4997768510389230	0,6250000000000000	0,7382424575738520	0,7689053178459950
hogfeat_128_16_4_9_pca	(800, 104)	crossvalidation	0,7136050711847340	0,6596683770726230	0,7058068483481920	0,7717777559844430
		holdout	0,7676892624945150	0,7071377408846500	0,7133097324362380	0,7444049585151980
hogfeat_128_32_2_9	(800, 325)	crossvalidation	0,5215734550703000	0,6270220761759760	0,7114460262557160	0,7713537269812650
		holdout	0,5648760858456820	0,6295585537074700	0,7378314393939390	0,7802465996569740
hogfeat_256_32_2_9	(800, 1765)	crossvalidation	0,4934979279611660	0,6694457395545480	0,7851698582336710	0,7737586068031630
		holdout	0,5336497959720940	0,6857017263427110	0,8000000000000000	0,7605179746429140
hogfeat_256_32_2_9_pca	(800, 94)	crossvalidation	0,7117194992366840	0,7070458495991490	0,7411259861031220	0,7891098109642110
		holdout	0,7118846153846150	0,7071352393252710	0,7616678611243820	0,7754959848842700
hogfeat_256_64_2_18	(800, 649)	crossvalidation	0,5898320688281290	0,6701022482748160	0,7357409305148180	0,7998504220055250
		holdout	0,6276256789187820	0,6833335063748490	0,7507843137254900	0,7621230158730150
hogfeat_256_64_2_9	(800, 325)	crossvalidation	0,5342477286812380	0,6922253620630580	0,7528476025759820	0,7913688830061230
		holdout	0,5421820405310970	0,6822132933003650	0,7757058823529410	0,7881657268170420
lbpfeat_256_12_96	(800, 99)	crossvalidation	0,6619546041645070	0,7210987441272380	0,7076606238058900	0,7477581742848490
		holdout	0,6754063134864820	0,6772592134699160	0,7442044334975370	0,7632450980392150
lbpfeat_256_24_192	(800, 195)	crossvalidation	0,6434402218884120	0,6899575899386750	0,6926714990383970	0,7331792347923360
		holdout	0,6738821138211380	0,7444012707722380	0,7300172510071390	0,7988491048593340
lbpfeat_256_6_48	(800, 51)	crossvalidation	0,6609864001873350	0,6778695328549210	0,7061809239555980	0,7618123780360840
		holdout	0,6385209669848220	0,7108805351275230	0,7320621884689350	0,7227600250626560
		Média	0,6086909267387500	0,6763788121452170	0,7374761745532400	0,7650328505118750
		Desvio Padrão	0,0891748838700598	0,0342800680968876	0,0310694625596781	0,0230171562073974
Friedman P Value:			KNN Config 2	Dtree Config 8	NB Config 2	MLP Config 10
0,000000000094850		KNN Config 2	1,0000000000000000	0,1492315671690420	0,0000734925136969	0,0000000034160632
		Dtree Config 8	0,1492315671690420	1,0000000000000000	0,0085326063520494	0,0000021527865066
		NB Config 2	0,0000734925136969	0,0085326063520494	1,0000000000000000	0,0208500581344520
		MLP Config 10	0,0000000034160632	0,0000021527865066	0,0208500581344520	1,0000000000000000

Comparativo Comites

dataset	shape	training_type	Bagging 12	Random Forest 12	Voting 1	Stacking 2
hogfeat_128_16_2_9	(800, 1765)	crossvalidation	0,7933272617171690	0,7663775120938670	0,7660369417792070	0,7869694216206520
		holdout	0,7676892624945150	0,7944046592734820	0,7083097165991900	0,7598202772404900
hogfeat_128_16_2_9_pca	(800, 118)	crossvalidation	0,7860869598095770	0,7508705305558300	0,7703000910113230	0,7660863672021340
		holdout	0,7731222917669710	0,7258627450980390	0,7616678611243820	0,7310476841396250
hogfeat_128_16_4_9	(800, 3601)	crossvalidation	0,7868971169174590	0,7368908157755460	0,7795784388359450	0,7966351132924620
		holdout	0,7691600985221670	0,7320598998082260	0,7605179746429140	0,7630235396000630
hogfeat_128_16_4_9_pca	(800, 104)	crossvalidation	0,7823477120713600	0,7349799844740320	0,7826902195356740	0,8062503365265780
		holdout	0,7752840909090900	0,7378314393939390	0,7923048048048040	0,7685759142596770
hogfeat_128_32_2_9	(800, 325)	crossvalidation	0,7635927173838570	0,7639974487471460	0,7690943522144390	0,7748276396041620
		holdout	0,7621230158730150	0,7819092207741380	0,7776720849060430	0,7810853242996940
hogfeat_256_32_2_9	(800, 1765)	crossvalidation	0,7866396768231800	0,8016952816080820	0,7597102266586600	0,7746617933213690
		holdout	0,7545420420420420	0,7507071024512880	0,7385750275386560	0,7370833333333330
hogfeat_256_32_2_9_pca	(800, 94)	crossvalidation	0,7873955984182630	0,7628011657620940	0,7894832034605460	0,8002625410158440
		holdout	0,7685759142596770	0,7731222917669710	0,7556396319408560	0,7627998737373730
hogfeat_256_64_2_18	(800, 649)	crossvalidation	0,8038175203183250	0,7927357958400620	0,7862555834548620	0,8075937264657320
		holdout	0,7731222917669710	0,8068355312217900	0,7598202772404900	0,7871626984126980
hogfeat_256_64_2_9	(800, 325)	crossvalidation	0,7913944985387460	0,7854247518565550	0,7760808123640680	0,7826573817429670
		holdout	0,7611333120204600	0,7942741935483870	0,7510184246112230	0,7611333120204600
lbpfeat_256_12_96	(800, 99)	crossvalidation	0,7332538459360240	0,7478638525763870	0,7365399286814240	0,7332238971900730
		holdout	0,7941157635467970	0,7816346567152510	0,6782147615502170	0,7813969222867520
lbpfeat_256_24_192	(800, 195)	crossvalidation	0,7452945471806790	0,7491618360472970	0,7377980973670330	0,7366951268372670
		holdout	0,7693410601234710	0,7818059628543500	0,7880984978876540	0,7444049585151980
lbpfeat_256_6_48	(800, 51)	crossvalidation	0,7603490938029830	0,7488199996451410	0,7279855466637730	0,7523276815967970
		holdout	0,7370897796530700	0,7819092207741380	0,7500000000000000	0,7556499355242070
		Média	0,7719039779956610	0,7659989957775850	0,7584746877030580	0,7688072833244000
		Desvio Padrão	0,0180937012554676	0,0241750995046157	0,0272835480459566	0,0227414153067510
Friedman P Value:						
0,488997847795918						



Conclusões

- O estudo demonstrou que **as técnicas supervisionadas aplicadas à classificação de imagens** apresentaram desempenho consistente e coerente com suas propriedades teóricas.
- Entre os classificadores individuais, o **MLP** obteve o **melhor desempenho médio ($F1 \approx 0,76$)** e estabilidade, confirmando sua capacidade de representar relações não lineares complexas.
- O **Naive Bayes** destacou-se pelo **baixo custo computacional** e bom equilíbrio entre simplicidade e acurácia.
- Entre os comitês, o **Bagging** apresentou o **melhor equilíbrio entre desempenho e estabilidade ($F1 \approx 0,77$)**, enquanto **Stacking** e **Random Forest** mantiveram resultados muito próximos, sem diferença estatística significativa.
- Os **descritores HOG** superaram os **LBP**, evidenciando maior poder discriminativo estrutural.
- A aplicação do **PCA** reduziu a dimensionalidade sem perda relevante de desempenho, otimizando o custo computacional.
- De modo geral, os **comitês aumentaram a estabilidade dos modelos**, mas os ganhos em desempenho foram marginais — indicando que, em bases moderadas, **modelos bem ajustados podem alcançar desempenho comparável a ensembles complexos**.

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